FACULTY OF ENGINEERING
AND ARCHITECTURE

Development, simulation & implementation of precoding techniques for satellite links

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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

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UNIVERSITY

Ghent, May 2026
Wannes Baes

Acknowledgements

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Ghent University

Faculty of Engineering and Architecture

Department of Telecommunications and Information Processing

Abstract — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

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I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

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Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not "Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

D. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

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E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [1].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
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[a]Sample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present

Fig. 1. Example of a figure caption.

them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization $\{A[m(1)]\}$", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks …". Instead, try "R. B. G. thanks…". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

[1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, "Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios," *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

Contents

List of Figures

Symbols and Notations

Abbreviations

BS	base station
CS	circularly symmetric
IBR	information bit rate
i.i.d.	independent and identically distributed
MI	mutual information
MIMO	multiple-input multiple-output
MU	multi-user
pdf	probability density function
RX	receiver
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
TX	transmitter
UT	user terminal
WSR	weighted sum rate

Mathematical Notations

TO DO

System Parameters

N_T Number of transmit antennas at the BS

N_R Number of receive antennas at each UT

N_S Number of data streams per UT

K Number of UTs

Symbols

TO DO

Chapter 1

Introduction

Part I

Background Material

Chapter 2

Communication Systems

This chapter establishes the system models and information-theoretic framework that underlie the analysis presented throughout this work. We begin by describing the general architecture of a digital communication system and then specialize to the system model adopted in this work. Finally, we review the fundamental information-theoretic concepts required to analyze the performance of these systems.

2.1 Digital Communication Systems

A general digital communication system transfers digital information from a source to a destination over a physical transmission medium, referred to as the channel. The corresponding block diagram is shown in Fig. 2.1.

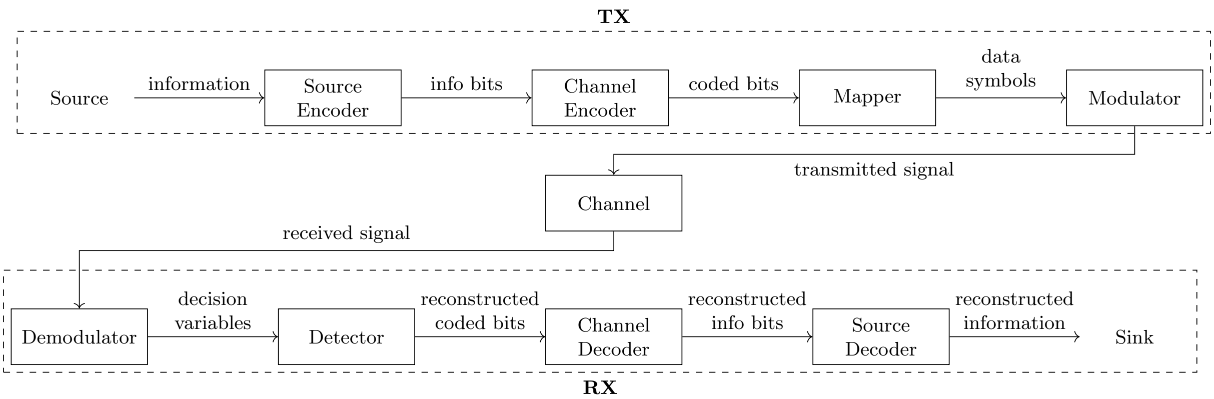


Figure 2.1: Block diagram of a general digital communication system.

At the transmitter (TX), the information sequence is first processed by a source encoder to remove redundancy, followed by a channel encoder that introduces controlled redundancy to enable error detection. The resulting coded bit sequence is partitioned into blocks of m_c bits, each of which is mapped onto a complex-valued symbol drawn from a constellation $\mathcal{X}$ of size 2^{m_c}. The modulator then converts the symbol sequence into a baseband signal suitable for transmission.

During transmission, the channel distorts the signal due to attenuation, multipath propagation, and additive noise, such that the received signal differs from the transmitted signal.

At the receiver (RX), the demodulator converts the received signal into a sequence of decision variables. These are processed by the detector to obtain estimates of the originally transmitted symbols, which are then mapped back to coded bits. Finally, the channel and source decoders reconstruct the original information sequence.

In this work, we consider a simplified version of this general architecture in order to focus on the aspects most relevant to our study. Source coding and decoding are omitted, and the information is assumed to be available as a sequence of independent and identically distributed (i.i.d.) bits without prior compression or digitization. Channel coding and decoding are also not considered, such that the analysis is restricted to uncoded transmission.

The transmission medium is assumed to be a wireless channel affected by fading and additive white Gaussian noise. Furthermore, the channel is considered to be frequency-flat, meaning that its response remains constant over the entire signal bandwidth. The specific channel models used in this work are introduced in the relevant chapters.

2.2 MIMO Communication Systems

When both the TX and the RX are equipped with multiple antennas, the system is referred to as a multiple-input multiple-output (MIMO) system.

Compared to a single-antenna link, the multiple antennas provide additional spatial degrees of freedom that can be exploited to increase data throughput, improve reliability, or achieve a trade-off between both. Spatial multiplexing techniques enable significant capacity gains by transmitting multiple independent data streams simultaneously over different antennas. A well-known example is the BLAST architecture [1]. Alternatively, spatial diversity can be achieved through space-time coding schemes, which distribute symbols across both space and time to improve robustness against fading. A classical example is the Alamouti scheme [2]. Throughout this work, the focus is on spatial multiplexing and, in particular, on how the simultaneous transmission of multiple data streams over a shared channel can be made most efficient through precoding.

2.2.1 SU-MIMO System Model

In the point-to-point case, where a single TX with N_T antennas communicates with a single RX with N_R antennas, the system is called a single-user (SU) MIMO system. A block diagram is shown in Fig. 2.2.

The discrete-time complex baseband input-output relation of a SU-MIMO Gaussian channel is characterized by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \tag{2.1}$$

where $\mathbf{x} \in \mathbb{C}^{N_T}$ is the transmitted signal vector, $\mathbf{y} \in \mathbb{C}^{N_R}$ is the received signal vector, $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix, and $\mathbf{n} \in \mathbb{C}^{N_R}$ is the additive noise vector. The (n_r, n_t)-th entry of $\mathbf{H}$ represents the complex channel coefficient between transmit antenna n_t and receive

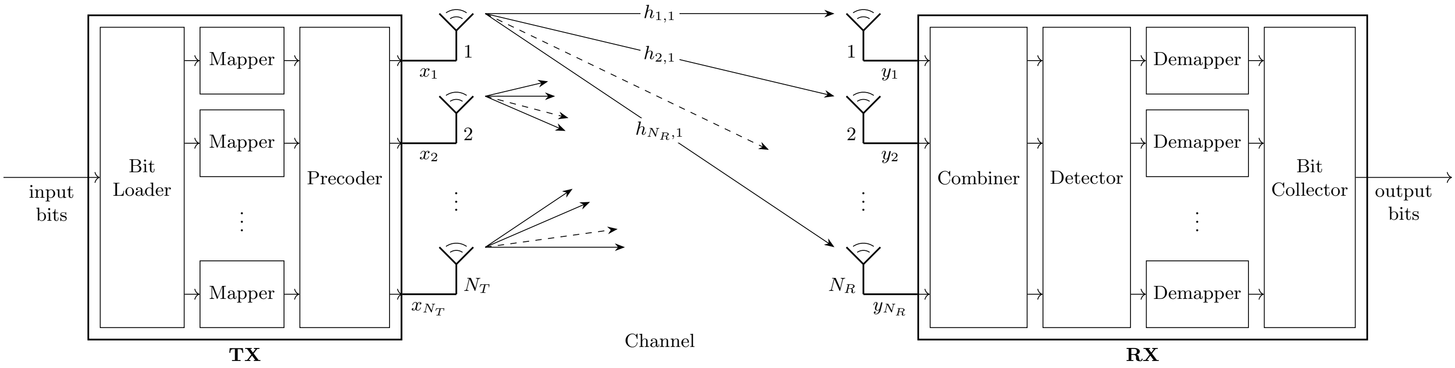


Figure 2.2: Block diagram of a SU-MIMO system.

antenna n_r. The noise vector is circularly symmetric (CS) complex Gaussian distributed, i.e. $\mathbf{n} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, N_0\,\mathbf{I}_{N_R})$, such that its entries are spatially independent, each with an expected power of N_0. The time index is omitted for brevity.

At the TX, the information bits are divided into N_S streams and mapped onto data symbols, which are collected in the vector $\mathbf{a} \in \mathbb{C}^{N_S \times 1}$. Then, the transmitted signal is obtained by mapping the N_S data symbol streams onto the N_T transmit antennas according to a linear precoding scheme, represented by a matrix $\mathbf{F} \in \mathbb{C}^{N_T \times N_S}$. The transmitted signal can therefore be expressed as

$$\mathbf{x} = \mathbf{F}\,\mathbf{a}. \tag{2.2}$$

The data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, such that $\mathbb{E}\big[\mathbf{a}\mathbf{a}^H\big] = \mathbf{I}_{N_S}$. Throughout this work, the power allocation is assumed to be incorporated into the precoding matrix, unless explicitly stated otherwise. Under this convention, the total transmit power is constrained by

$$\mathbb{E}\big[\|\mathbf{x}\|^2\big] = \mathrm{tr}\big(\mathbf{F}^H\mathbf{F}\big) \leq P_t. \tag{2.3}$$

At the RX, the N_R received signals are linearly combined to obtain the decision-variable vector $\mathbf{z} \in \mathbb{C}^{N_S \times 1}$. The combiner is represented by a matrix $\mathbf{G} \in \mathbb{C}^{N_S \times N_R}$. The decision variables can therefore be expressed as

$$\mathbf{z} = \mathbf{G}\,\mathbf{y}. \tag{2.4}$$

Based on $\mathbf{z}$, the detector forms estimates of the transmitted data symbols, which are then mapped back to bits.

Substituting (2.2) into (2.1) and applying (2.4) yields the relation between the transmitted data symbols and the decision variables:

$$\mathbf{z} = (\mathbf{G}\,\mathbf{H}\,\mathbf{F})\ \mathbf{a} + \mathbf{G}\,\mathbf{n}, \tag{2.5a}$$

$$= \mathbf{T}\,\mathbf{a} + \mathbf{n}_{\mathbf{G}}, \tag{2.5b}$$

where $\mathbf{T} \in \mathbb{C}^{N_S \times N_S}$ is the MIMO transfer matrix and $\mathbf{n}_{\mathbf{G}}$ represents spatially correlated Gaussian noise. It is evident that, whenever the transfer matrix is not diagonal, the data symbols of different streams interfere with each other. The design of the precoding and combining matrices is therefore essential to suppress this inter-stream interference and improve overall system performance.

2.2.2 MU-MIMO System Model

When multiple receivers are served simultaneously by a single multi-antenna transmitter, the system is referred to as a multi-user (MU) MIMO system. In this context, the transmitter is called the base station (BS) and the individual receivers are referred to as user terminals (UTs). A block diagram is shown in Fig. 2.3.

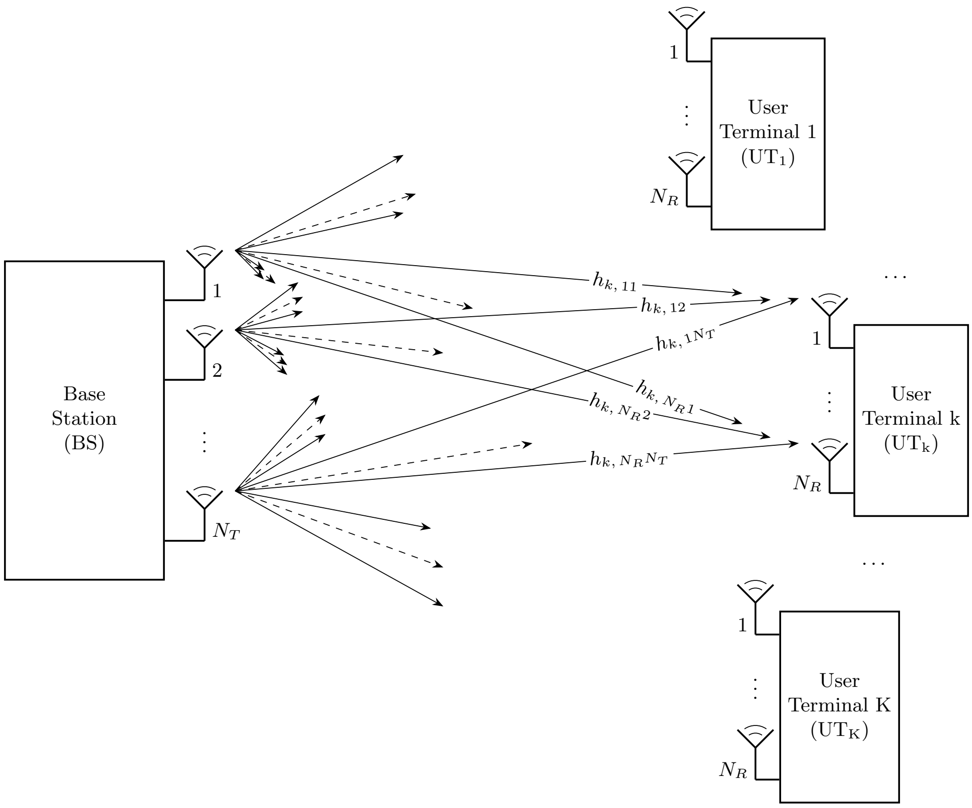


Figure 2.3: Block diagram of a MU-MIMO system.

Throughout this work, the BS is assumed to be equipped with more antennas than all UTs combined, and each UT is assumed to receive at most as many data streams as it has receive antennas. For notational convenience, all UTs are assumed to receive the same number of data streams, denoted by N_S. In practical scenarios where this does not hold, N_S is defined as the maximum number of data streams intended for any single UT. The data symbol vectors intended for a UT receiving fewer streams are padded with zeros.

Analogous to the SU-MIMO case, the communication between the BS and a certain $\mathrm{UT_k}$ at a particular time instant is characterized by

$$\mathbf{y}_k = \mathbf{H}_k \, \mathbf{x} + \mathbf{n}_k, \qquad k = 1, \ldots, K, \tag{2.6}$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix that describes the link between the BS and $\mathrm{UT_k}$.

The transmitted signal vector $\mathbf{x} \in \mathbb{C}^{N_T}$ is now a function of the data symbol vectors intended for different UTs. The BS assigns a dedicated precoding scheme $\mathbf{F}_k \in \mathbb{C}^{N_T \times N_S}$ to the data symbol vector $\mathbf{a}_k$ intended for $\mathrm{UT_k}$, and transmits the superposition of all precoded streams:

$$\mathbf{x} = \sum_{k=1}^{K} \mathbf{F}_k \, \mathbf{a}_k. \tag{2.7}$$

Since the data symbols intended for different UTs are assumed to be uncorrelated, the total transmit power constraint can now be formulated as

$$\mathbb{E}\big[\|\mathbf{x}\|^2\big] = \sum_{k=1}^{K} \mathrm{tr}\big(\mathbf{F}_k^H \mathbf{F}_k\big) \leq P_t. \tag{2.8}$$

Applying the combining scheme $\mathbf{G}_k \in \mathbb{C}^{N_S \times N_R}$ at $\mathrm{UT_k}$ to the received signal vector and substituting (2.7) into (2.6) yields

$$\mathbf{z}_k = \underbrace{(\mathbf{G}_k \, \mathbf{H}_k \, \mathbf{F}_k) \;\, \mathbf{a}_k}_{\text{desired signal}} + \underbrace{\sum_{k'=1,\, k' \neq k}^{K} (\mathbf{G}_k \mathbf{H}_k \mathbf{F}_{k'}) \;\, \mathbf{a}_{k'}}_{\text{inter-user interference}} + \underbrace{\mathbf{G}_k \, \mathbf{n}_k}_{\text{noise}}. \tag{2.9}$$

The key difference from the SU-MIMO scenario is the presence of inter-user interference. Each UT receives not only its desired signal but also the signals intended for all other UTs. Suppressing this interference through precoder design at the BS is an important challenge in MU-MIMO systems.

The overall input-output relation of the MU-MIMO Gaussian channel can be written compactly as

$$\mathbf{z} = (\mathbf{G} \, \mathbf{H} \, \mathbf{F}) \;\, \mathbf{a} + \mathbf{G} \, \mathbf{n}, \quad \text{with} \tag{2.10}$$

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_K \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{G}_1 & & \\ & \ddots & \\ & & \mathbf{G}_K \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_K \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_1 & \dots & \mathbf{F}_K \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_K \end{bmatrix}.$$

where $\mathbf{G} \in \mathbb{C}^{KN_S \times KN_R}$ denotes the compound combining matrix, $\mathbf{H} \in \mathbb{C}^{KN_R \times N_T}$ the compound channel matrix, and $\mathbf{F} \in \mathbb{C}^{N_T \times KN_S}$ the compound precoding matrix.

2.3 Information Theory Concepts

In this section, we introduce the information-theoretic concepts needed for the analysis of MIMO communication systems. We first discuss mutual information, then define the achievable rate and capacity for SU-MIMO channels, and finally extend these concepts to the MU scenario.

2.3.1 Mutual Information

Consider a single use of a SU-MIMO Gaussian channel, as described by (2.1). The transmitter generates a random vector $\mathbf{x} \in \mathbb{C}^{N_T}$ according to the input distribution with probability density function (pdf) $p(\mathbf{x})$, while the receiver observes the random vector $\mathbf{y} \in \mathbb{C}^{N_R}$. The relation between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ is probabilistic and is fully characterized by the channel law, described by the conditional pdf $p(\mathbf{y} \mid \mathbf{x})$.

The number of information bits transmitted per channel use is referred to as the information bit rate (IBR). The mutual information (MI) between $\mathbf{x}$ and $\mathbf{y}$ quantifies how much information the received vector $\mathbf{y}$ provides about the transmitted vector $\mathbf{x}$. In other words, it represents the average reduction in the uncertainty about $\mathbf{x}$ obtained from observing $\mathbf{y}$. It is given by

$$I(\mathbf{x};\mathbf{y}) = \mathbb{E}\left[\log_2\left(\frac{p(\mathbf{y} \mid \mathbf{x})}{p(\mathbf{y})}\right)\right], \tag{2.11}$$

where the expectation is taken with respect to the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ and $p(\mathbf{y})$ is the marginal distribution of $\mathbf{y}$.

According to Shannon's channel coding theorem [3], reliable communication is possible if the transmission IBR is strictly smaller than the MI between $\mathbf{x}$ and $\mathbf{y}$. Here, reliable communication means that the decoding error probability can be made arbitrarily small by using sufficiently long channel codes.

2.3.2 Achievable Rate and Capacity

For a given input distribution $p(\mathbf{x})$ and channel law $p(\mathbf{y} \mid \mathbf{x})$, any IBR r smaller than the MI is said to be achievable. The MI is thus a supremum (least upper bound) on the achievable rate.

The capacity C of a SU-MIMO channel is the largest achievable rate. Equivalently, it is the highest spectral efficiency for which reliable communication is possible. It is defined as the maximum MI between $\mathbf{x}$ and $\mathbf{y}$ over all possible input distributions $p(\mathbf{x})$:

$$C = \max_{p(\mathbf{x})} I(\mathbf{x};\mathbf{y}). \tag{2.12}$$

Typically, the maximization is subject to the transmit power constraint, as formulated in (2.3).

In [4], the capacity for a SU-MIMO Gaussian communication system is derived. First, it is shown that under the total transmit power constraint, the MI is maximized in case of Gaussian signaling, i.e. the channel input $\mathbf{x}$ is CS complex Gaussian distributed. Then, the mutual information between $\mathbf{x}$ and $\mathbf{y}$ for a given covariance matrix $\mathbf{C}_\mathrm{x}$ is obtained as

$$I(\mathbf{x};\mathbf{y}) = \log_2\left(\det\left(\mathbf{I}_{N_R} + \mathbf{C}_\mathrm{n}^{-1}\,\mathbf{H}\,\mathbf{C}_\mathrm{x}\,\mathbf{H}^\mathrm{H}\right)\right). \tag{2.13}$$

The capacity then results from maximizing (2.13) over all positive semidefinite covariance matrices $\mathbf{C}_{\mathrm{x}}$ that satisfy the transmit power constraint:

$$\begin{aligned} C &= \max_{\mathbf{C}_{\mathrm{x}}} \log_2\left(\det\left(\mathbf{I}_{N_R} + \mathbf{C}_{\mathrm{n}}^{-1}\,\mathbf{H}\,\mathbf{C}_{\mathrm{x}}\,\mathbf{H}^{\mathrm{H}}\right)\right) \\ \text{s.t.} \quad & \mathbf{C}_{\mathrm{x}} \succeq 0, \quad \mathrm{tr}(\mathbf{C}_{\mathrm{x}}) \leq P_t. \end{aligned} \tag{2.14}$$

2.3.3 Rate Region and Sum Capacity

In contrast to the SU scenario, where the IBR is a scalar r, a MU-MIMO channel is characterized by a rate tuple $(r_1, \ldots, r_K)$, where each element r_k denotes the IBR of a single UT. The rate region $\mathcal{R}$ is the set of all rate tuples whose individual IBRs are simultaneously achievable under the given system constraints. The rate region is a convex set, since any convex combination of achievable rate tuples is itself achievable through time-sharing between the corresponding transmission strategies. An example rate region for $K = 2$ users is shown in Fig. 2.4.

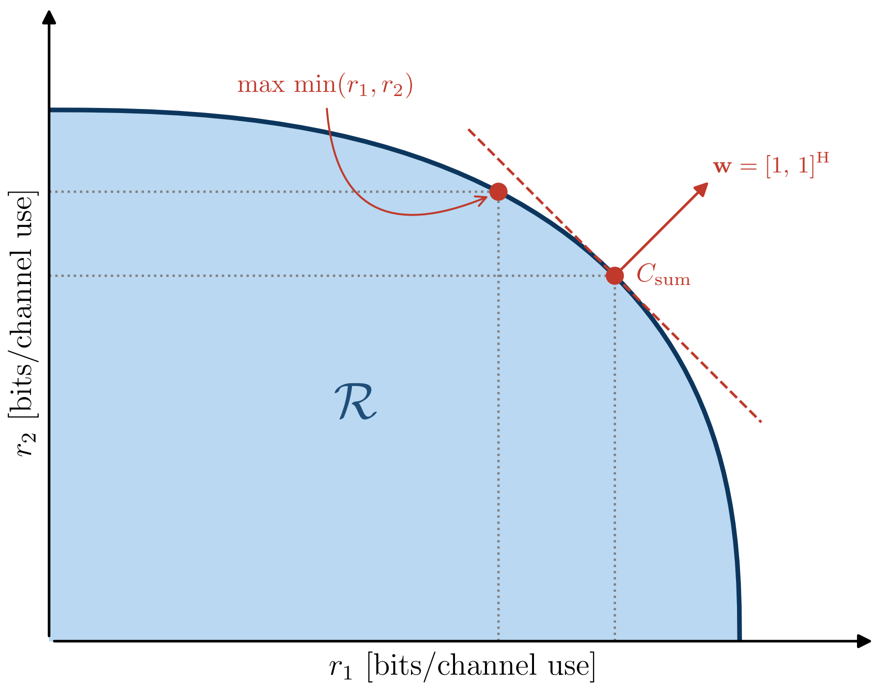


Figure 2.4: Example of a rate region $\mathcal{R}$ for a MU-MIMO system with $K = 2$ users. The sum capacity point is the boundary point whose outward normal is the all-ones vector. The max–min point is given by the intersection of the boundary with the line $r_1 = r_2$. It corresponds to a fair operating point where both users achieve the same IBR.

The rate region captures the trade-offs inherent to multi-user transmission: increasing the IBR of one user may reduce the rates achievable by others due to interference and shared resources. It is therefore desirable to operate on the boundary of the rate region, where no user's rate can be increased without reducing that of another.

The sum capacity C_{sum} is the maximum achievable sum rate (SR), i.e., the sum of all individual IBRs:

$$C_{\text{sum}} = \max_{(r_1,\ldots,r_K)\in\mathcal{R}} \sum_{k=1}^{K} r_k. \tag{2.15}$$

More generally, we define the weighted sum rate (WSR) as

$$\text{WSR} = \sum_{k=1}^{K} w_k\, r_k, \tag{2.16}$$

where $w_k \geq 0$ is the weight assigned to UT_k. The full boundary of the rate region can then be traced by maximizing the WSR over all achievable rate tuples for varying weight vectors $\mathbf{w} = (w_1, \ldots, w_K)$. The sum capacity corresponds to the special case where all weights are equal to one. The maximization in both cases is typically subject to the total transmit power constraint (2.8).

In [5, 6, 7], the sum capacity of the MU-MIMO Gaussian channel is derived by exploiting the uplink–downlink duality: the downlink broadcast channel and the uplink multiple-access channel yield the same rate region under the same total transmit power constraint. It therefore suffices to solve the uplink problem, which is more tractable.

In the uplink channel, all K UTs transmit simultaneously, with UT_k sending a vector $\mathbf{x}_k^{\text{up}} \in \mathbb{C}^{N_R}$ to the BS. The MI between the transmitted vectors $\mathbf{x}_1^{\text{up}}, \ldots, \mathbf{x}_K^{\text{up}}$ and the received vector $\mathbf{y}$, given input covariance matrices $\mathbf{C}_{\text{x},k}$, is defined as

$$I(\mathbf{x}_1^{\text{up}}, \ldots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \mathbb{E}\left[\log_2\left(\frac{p(\mathbf{y} \mid \mathbf{x}_1^{\text{up}}, \ldots, \mathbf{x}_K^{\text{up}})}{p(\mathbf{y})}\right)\right], \tag{2.17}$$

where the expectation is taken with respect to the joint pdf of $(\mathbf{x}_1^{\text{up}}, \ldots, \mathbf{x}_K^{\text{up}})$ and $\mathbf{y}$, and $p(\mathbf{y})$ is the marginal pdf of $\mathbf{y}$. It is further obtained as

$$I(\mathbf{x}_1^{\text{up}}, \ldots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \log_2\left(\det\left(\mathbf{I}_{N_T} + \mathbf{C}_\text{n}^{-1} \sum_{k=1}^{K} \mathbf{H}_k\, \mathbf{C}_{\text{x},k}^{\text{up}}\, \mathbf{H}_k^{\text{H}}\right)\right). \tag{2.18}$$

The uplink SR is achievable if and only if the sum of the IBRs of all UTs is strictly smaller than the MI as given in (2.18). Therefore, the sum capacity of the downlink channel is obtained by maximizing (2.18) over all positive semidefinite covariance matrices satisfying the total transmit power constraint and applying the duality result:

$$\begin{aligned} C_{\text{sum}} = &\max_{\mathbf{C}_{\text{x},1}^{\text{up}},\ldots,\mathbf{C}_{\text{x},K}^{\text{up}}} \log_2\left(\det\left(\mathbf{I}_{N_T} + \mathbf{C}_\text{n}^{-1} \sum_{k=1}^{K} \mathbf{H}_k\, \mathbf{C}_{\text{x},k}^{\text{up}}\, \mathbf{H}_k^{\text{H}}\right)\right) \\ \text{s.t.} \quad &\sum_{k=1}^{K} \text{tr}\left(\mathbf{C}_{\text{x},k}^{\text{up}}\right) \leq P_t \quad \text{and} \quad \mathbf{C}_{\text{x},k}^{\text{up}} \succeq 0 \quad \forall k = 1, \ldots, K. \end{aligned} \tag{2.19}$$

Chapter 3

Precoding for SU-MIMO systems

3.1 Capacity Achieving Precoding

Consider the SU-MIMO system model introduced in Section 2.2.1 together with the associated capacity optimization problem in (2.14). We first derive a more explicit expression for the channel capacity and then show that this capacity can be achieved by means of linear precoding. Parts of this section follow a structure similar to that in Chapter 3 of [8].

3.1.1 SU-MIMO Capacity

The singular value decomposition (SVD) [9] of the channel matrix yields

$$\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^H, \tag{3.1}$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times N_R}$ and $\mathbf{V} \in \mathbb{C}^{N_T \times N_T}$ are unitary matrices, and $\boldsymbol{\Sigma} \in \mathbb{R}^{N_R \times N_T}$ is a rectangular diagonal matrix which contains the singular values of $\mathbf{H}$ on its main diagonal in decreasing order. The columns of $\mathbf{U}$ and $\mathbf{V}$ contain the left and right singular vectors of $\mathbf{H}$, respectively.

Now define the transformed variables $\tilde{\mathbf{y}} = \mathbf{U}^H\mathbf{y}$, $\tilde{\mathbf{x}} = \mathbf{V}^H\mathbf{x}$, and $\tilde{\mathbf{n}} = \mathbf{U}^H\mathbf{n}$. Since multiplication by a unitary matrix preserves the distribution of CS complex Gaussian noise, $\tilde{\mathbf{n}}$ has the same distribution as $\mathbf{n}$. Moreover, because $\mathbf{U}$ and $\mathbf{V}$ are unitary, the transformations between $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ are bijective. The original channel from (2.1) can therefore be represented equivalently as

$$\tilde{\mathbf{y}} = \boldsymbol{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \tag{3.2}$$

which is obtained by applying the SVD of the channel matrix and performing a change of variables (see Appendix B). The channel in (3.2) is equivalent to the original channel in the sense that the transformation preserves the mutual information between input and output. Consequently, both channels have the same capacity.

The channel matrix $\mathbf{H}$ has exactly as many non-zero singular values $\{\sigma_i\}$ as its rank, denoted as R_H. Written component-wise, the equivalent channel becomes

$$\tilde{y}_i = \sigma_i \, \tilde{x}_i + \tilde{n}_i \qquad \text{for} \quad i = 1, \ldots, R_H, \tag{3.3}$$

with $\tilde{y}_i = \tilde{n}_i$ for $R_H < i \leq N_R$. The equivalent channel decomposes into R_H independent parallel scalar Gaussian subchannels, each with its own channel gain σ_i. These subchannels and their gains are referred to as eigen subchannels and eigenmodes, respectively. Since the components $\tilde{x}_i$ for $R_H < i \leq N_T$ have no effect on the output, they carry no information and can be set to zero.

For each eigen subchannel, the mutual information (MI) is maximised by a zero-mean, CS complex Gaussian input [4]. Under this choice, the optimisation problem (2.14) simplifies to

$$\begin{aligned} C = & \max_{\{\mathbb{E}[|\tilde{x}_i|^2]\}} \sum_{i=1}^{R_H} \log_2\left(1 + \mathbb{E}\left[|\tilde{x}_i|^2\right] \frac{\sigma_i^2}{N_0}\right) \\ \text{s.t.} & \quad \sum_{i=1}^{R_H} \mathbb{E}\left[|\tilde{x}_i|^2\right] \leq P_t. \end{aligned} \tag{3.4}$$

This maximization problem is solved using the Lagrange multiplier method (see Appendix **??**). The variances of the channel input need to be chosen according to the waterfilling solution [10]:

$$\mathbb{E}\left[\|\tilde{x}_i\|^2\right] = \left(\mu_{\mathrm{w}} - \frac{N_0}{\sigma_i^2}\right)^+, \tag{3.5}$$

where the water level μ_{w} is found iteratively to satisfy the transmit power constraint with equality.

The resulting capacity is given by

$$C = \sum_{i=1}^{R_H} \log_2\left(\max(1, \, \mu_{\mathrm{w}} \frac{\sigma_i^2}{N_0})\right). \tag{3.6}$$

3.1.2 Optimal Precoder Design

In this section, we design a linear precoder and combiner that together achieve the capacity of the SU-MIMO channel. We assume that the data symbol vector is CS complex Gaussian distributed and spatially uncorrelated, and restrict ourselves to the case where the number of data streams is not fixed in advance, but can be chosen freely as part of the design.

For an arbitrary precoding matrix $\mathbf{F}$ and combining matrix $\mathbf{G}$, the data processing inequality [11] yields $I(\mathbf{x}; \mathbf{y}) \geq I(\mathbf{a}; \mathbf{z})$, where the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ is given by:

$$I(\mathbf{a}; \mathbf{z}) = \log_2\big(\det\big(\mathbf{I}_{N_s} + (\mathbf{G}\,\mathbf{C}_{\mathrm{n}}\,\mathbf{G}^H)^{-1}\,(\mathbf{GHF})\,\mathbf{C}_{\mathrm{a}}\,(\mathbf{GHF})^H\big)\big) \tag{3.7a}$$

$$= \log_2\left(\det\left(\mathbf{I}_{N_s} + \tfrac{1}{N_0}(\mathbf{G}\mathbf{G}^H)^{-1}\,(\mathbf{GHF})\,(\mathbf{GHF})^H\right)\right). \tag{3.7b}$$

Expression (3.7a) follows directly from (2.13) and the structural similarity between (2.1) and (2.5). Expression (3.7b) then follows from (3.7a) upon substituting $\mathbf{C}_\mathrm{a} = \mathbf{I}_{N_s}$ and $\mathbf{C}_\mathrm{n} = N_0 \mathbf{I}_{N_R}$.

Motivated by the capacity derivation in the previous subsection, we choose the columns of $\mathbf{F}$ and the rows of $\mathbf{G}$ as the R_H most dominant right and left singular vectors of $\mathbf{H}$, respectively. These are precisely the singular vectors associated with the non-zero singular values of $\mathbf{H}$. The number of data streams is therefore chosen equal to the rank of the channel matrix. In addition, the precoder incorporates the power allocation through a diagonal matrix $\mathbf{P}$. More specifically, we set

$$\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \qquad \text{and} \qquad \mathbf{G} = \mathbf{U}_{N_s}^H. \tag{3.8}$$

It follows from (3.8) that $I(\mathbf{a};\mathbf{z}) = I(\mathbf{x};\mathbf{y})$ for an appropriate choice of the power allocation matrix $\mathbf{P}$. This result can be verified in a brute-force way by substituting the particular precoder and combiner matrices from (3.8) into (3.7) and replacing the channel matrix by its SVD (see Appendix B).

An alternative and more elegant approach consists in substituting the particular precoder and combiner into (2.5) and comparing the resulting expression with the equivalent channel given in (3.2). This yields

$$\begin{aligned} \mathbf{z} &= \mathbf{U}_{N_s}^H\, \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^H\, \mathbf{V}_{N_s} \sqrt{\mathbf{P}}\, \mathbf{a} + \mathbf{U}_{N_s}^H \mathbf{n} && \text{(3.9a)} \\ &= \boldsymbol{\Sigma}_{N_s} \sqrt{\mathbf{P}}\, \mathbf{a} + \tilde{\mathbf{n}}, && \text{(3.9b)} \end{aligned}$$

where $\boldsymbol{\Sigma}_{N_s} \in \mathbb{R}^{N_s \times N_s}$ contains the R_H non-zero singular values of $\mathbf{H}$ on its diagonal. Comparing (3.9) with (3.2), one can readily verify that $I(\mathbf{a};\mathbf{z}) = I(\mathbf{x};\mathbf{y})$, provided that the power allocated to each stream is set according to the waterfilling solution. Hence, transmission with the precoder and combiner from (3.8) achieves the capacity of the SU-MIMO channel.

Denoting the sth element of the decision variable vector $\mathbf{z}$ by z_s, one can write (3.9) componentwise as

$$z_s = \sigma_s \sqrt{p_s}\, a_s + \tilde{n}_s \qquad \text{for} \quad s = 1, \ldots, N_s. \tag{3.10}$$

This expression shows that the precoder and combiner in (3.8) diagonalize the SU-MIMO transfer matrix from (2.5). The original SU-MIMO channel is decoupled into $N_s = R_H$ independent parallel scalar single-input single-output (SISO) Gaussian subchannels with gains σ_s, which coincides exactly with the decomposition of the equivalent channel in (3.3). A visualization of this decomposition is shown in Fig. 3.1.

As a result, the individual data streams become statistically independent. Detection therefore reduces to symbol-by-symbol processing rather than joint detection of the full decision-variable vector. This leads to a significant reduction in receiver complexity, especially for systems with a large number of antennas and data streams.

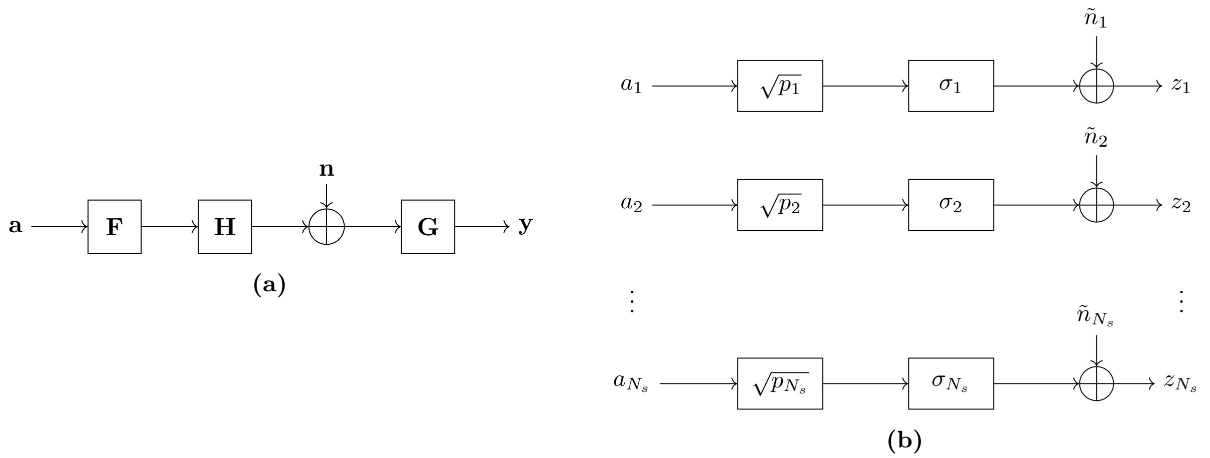


Figure 3.1: A block diagram of the original SU-MIMO channel (a) and its decomposition into parallel SISO Gaussian eigen subchannels (b) by means of the optimal precoder and combiner.

No general closed-form expression exists for the optimal water level μ_{w}, and hence for the resulting power allocation. Nevertheless, the optimal power allocation is fully characterized by

$$\sum_{i=1}^{N_s} p_i = P_t \qquad \text{and} \qquad p_i = \left(\mu_{\mathrm{w}} - \frac{N_0}{\sigma_i^2}\right)^+ . \tag{3.11}$$

Since the total allocated power is a monotonically increasing function of the water level, the latter can be determined numerically by means of a bisection method. A more efficient approach exploits the fact that the number of active streams (those receiving strictly positive power) is a discrete quantity. First, the number of active streams is determined via a binary search in at most $\lceil \log_2 N_s \rceil$ iterations, after which the water level follows in closed form. Finally, the power allocated to each stream is computed. The complete procedure is summarized in Algorithm 1, in Appendix **??**. Note that this approach relies on the fact that the set of candidate streams is finite and therefore does not extend in the same form to waterfilling algorithms for power allocation across the frequency spectrum of a frequency-selective channel.

The resulting solution admits a natural waterfilling interpretation, which also motivates the name of the algorithm. For each eigen subchannel, the quantity $\gamma_i^{-1} = N_0/\sigma_i^2$ can be interpreted as a floor level, while the water level μ_{w} is shared across all subchannels. Power is poured into each eigen subchannel container, so to speak, until the common water level is reached. Consequently, no power is assigned to eigen subchannels whose floor level exceeds the water level. The structure of the optimal power allocation is illustrated in Fig. 3.2 for a SU-MIMO system with four transmit and four receive antennas operating at an signal-to-noise ratio (SNR) of 5 dB: eigen subchannels with stronger channel conditions, i.e. larger gains σ_i, receive more power than weaker ones.

At extremely low SNR, the waterfilling solution corresponds to the eigenbeamforming power allocation strategy, which concentrates all available power on the strongest eigen subchannel. As a result, only a single data stream is transmitted in this SNR regime. This behavior is illustrated in Fig. 3.3a. Conversely, at an extremely high SNR, the optimal power allocation strategy reduces to the uniform power allocation, which distributes power equally across all eigen subchannels,

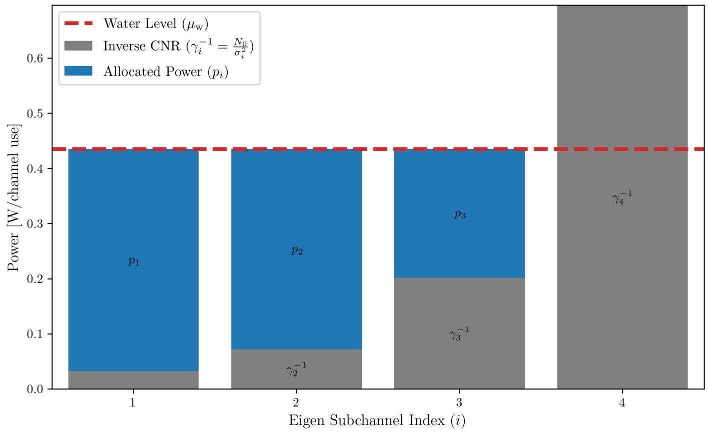
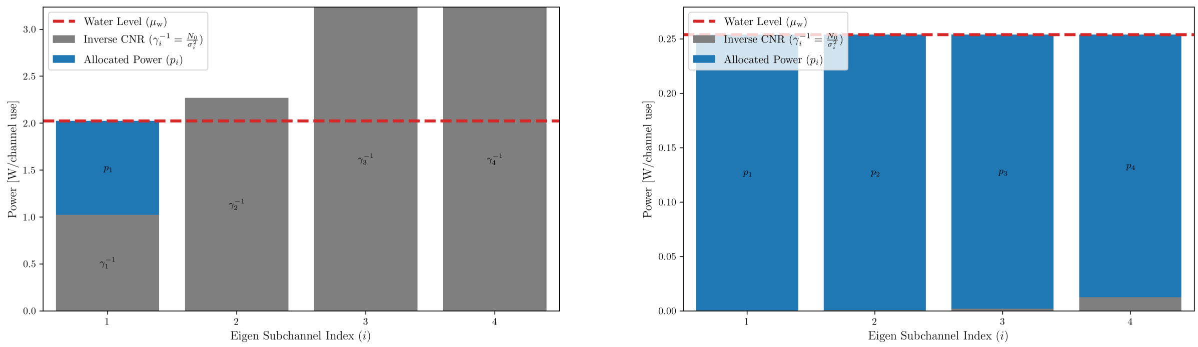


Figure 3.2: Optimal power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas and a total transmit power P_t of 1 W per channel use. The system operates at an SNR of 5 dB.

as illustrated in Fig. 3.3b. It is worth noting that uniform power allocation does not require knowledge of the current channel matrix, in contrast to the general waterfilling solution.



(a) SNR $= -10$ dB

(b) SNR $= 25$ dB

Figure 3.3: Waterfilling power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas at different SNR extremes. The maximum transmit power P_t is 1 W per channel use.

3.2 Joint Design for Linear Precoder and Combiner

3.2.1 Generalized WMMSE Design

3.2.2 Maximum Information Rate Design

3.2.3 Quality of Service Design

Part II

Precoding for Satellite Communications

Part III

Appendices

Appendix A

Waterfilling for Power Allocation

In this appendix, we solve the problem of optimal power allocation accross independent scaler Gaussian channels using the method of Lagrangian multipliers. The key principles of the method are explained alongside the derivation. This method is also employed to solve other optimization problems encountered in this dissertation. The solution to the optimal power allocation problem gives rise to the waterfilling algorithm, which is discussed at the end of this appendix.

A.1 Optimal Power Allocation using Lagrangian Multipliers

Consider N independent SISO Gaussian trasmission channels. Optimal power allocation across these channels can be formulated as a constrained maximization problem. The objective is to distribute the total available transmit power among the different channels such that the combined achievable rate is maximized.

Consider the following general optimization problem:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^{N} \alpha_n \log_2 (1 + \gamma_n \, p_n) \\ \text{s. t.} \quad & \sum_{n=1}^{N} p_n = P_t \quad \text{and} \quad p_n \geq 0 \quad \forall n = 1, \ldots, N \end{aligned} \tag{A.1}$$

where p_n denotes the power allocated to channel n, and P_t denotes the total available transmit power. The factors $\{\alpha_n\}$ represent weighting factors that allow for unequal prioritization of the individual transmission channels, while $\{gamma_n\}$ capture the ratio between the channel gain and the noise power of each indivudual channel.

To solve the constrained optimization problem in (A.1), the method of Lagrangian multipliers is employed. The core idea of this method is to convert a constrained optimization problem into an unconstrained one by absorbing the constraints into a modified objective function, known as the Lagrangian. This is achieved by introducing auxiliary variables, called Lagrange multipliers, one for each constraint, which penalize violations of the constraints.

Since the optimization problem in (A.1) involves both an equality constraint (total power budget) and inequality constraints (non-negativity of the individual power allocations), both types must be incorporated into the Lagrangian. Equality constraints enter with an unconstrained multiplier $\mu \in \mathbb{R}$, whereas inequality constraints of the form $p_n \geq 0$ are rewritten as $-p_n \leq 0$ and enter with a non-negative multiplier $\lambda_n \geq 0$, ensuring that the penalty is only active when the constraint is binding. The resulting Lagrangian is

$$\mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = \sum_{n=1}^{N} \alpha_n \log_2 (1 + \gamma_n p_n) - \mu \left(\sum_{n=1}^{N} p_n - P_t \right) - \sum_{n=1}^{N} \lambda_n p_n \tag{A.2}$$

where μ is the Lagrange multiplier associated with the total power constraint, and λ_n are the multipliers associated with the non-negativity constraints on the individual power allocations.

Since the objective function is concave and all constraints are affine, the optimization problem is convex. For convex problems, the Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient for a solution to be globally optimal [12, 13]. These conditions are given by

1. Stationarity: $\nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0$
2. Primal feasibility: $p_n \geq 0 \quad \text{and} \quad \sum_{n=1}^{N} p_n = P_t$
3. Dual feasibility: $\lambda_n \geq 0$
4. Complementary slackness: $\begin{cases} p_n > 0 \implies \lambda_n = 0 \\ p_n = 0 \implies \lambda_n \geq 0 \end{cases}$

From the KKT conditions, we can derive the optimal power allocation for each SISO channel. The derivation proceeds by successively applying the stationarity, complementary slackness, and primal feasibility conditions.

The stationarity condition requires that the gradient of the Lagrangian with respect to each optimization variable vanishes at the optimum. This expresses the balance between the marginal gain in the achievable rate and the penalties introduced by the power constraints. We find:

$$\nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad \iff \quad \frac{\partial \mathcal{L}}{\partial p_n} = \frac{\alpha_n \gamma_n}{\ln(2)\,(1 + \gamma_n p_n)} - \mu - \lambda_n = 0, \quad \forall n = 1, \ldots, N \tag{A.4}$$

The complementary slackness condition links the non-negativity constraint on the allocated power to its associated Lagrange multiplier. When a positive amount of power is allocated to channel n, the corresponding multiplier λ_n is zero, and the stationarity condition reduces to:

$$\frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu = 0 \tag{A.5a}$$

$$\iff \quad p_n = \frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \tag{A.5b}$$

The primal feasibility condition enforces the non-negativity of the power allocations. As a result, the above expression for p_n must be clipped to zero whenever it becomes negative:

$$p_n = \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)^+ \tag{A.6}$$

Finally, the primal feasibility condition also requires the total allocated power to equal the available transmit power P_t. By substituting the expression for p_n from Eq. (**??**) into this constraint, we obtain an implicit equation for the the Lagragian multiplier μ:

$$\sum_{n=1}^{N} \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)^+ = P_t \tag{A.7}$$

Since μ cannot be obtained in closed form, it is determined iteratively such that the total allocated power satisfies (A.7). Owing to the monotonic relationship between μ and the total allocated power, a bisection method is typically employed to efficiently compute its optimal value.

A.2 Waterfilling Algorithm

Applied to the more specific scenario considered in Section 3.1, no prioritization is employed, i.e. $\alpha_n = 1$, and the SISO Gaussian channels correspond to the eigen subchannels.

Algorithm 1 Waterfilling Algorithm for power allocation across eigen subchannels

Require: The total transmit power P_t; the SNR ratios $\gamma_i = \sigma_i / N_0$ for $i = 1, \dots, N_s$, sorted in decreasing order.

Ensure: The optimal power allocation p_i for $i = 1, \dots, N_s$.

Step 1: Determine the optimal number of active streams

N_s^{LB}, $N_s^{\mathrm{UB}} \leftarrow 0$, N_s

while $N_s^{\mathrm{UB}} - N_s^{\mathrm{LB}} > 1$ **do**

  Update the number of active streams:

  $N_s^{\mathrm{iter}} \leftarrow \left\lceil (N_s^{\mathrm{UB}} + N_s^{\mathrm{LB}})/2 \right\rceil$

  Compute the minimum required power to activate N_s^{iter} streams:

  $P_{\min} \leftarrow \sum_{i=1}^{N_s^{\mathrm{iter}}} \left(1/\gamma_{N_s^{\mathrm{iter}}} - 1/\gamma_i \right)$

  Update the bound on the number of active streams:

  if $P_{\min} \leq P_t$ **then** $N_s^{\mathrm{LB}} \leftarrow N_s^{\mathrm{iter}}$ **else** $N_s^{\mathrm{UB}} \leftarrow N_s^{\mathrm{iter}}$

end while

Step 2: Compute the power allocated to each stream

Fill the N_s^{LB} active stream containers:

  $p_i \leftarrow 1/\gamma_{N_s^{\mathrm{LB}}} - 1/\gamma_i \quad$ for $i = 1, \dots, N_s^{\mathrm{LB}}$

Distribute the remaining power equally among the active streams:

  $p_i \leftarrow p_i + \left(P_t - \sum_{i=1}^{N_s^{\mathrm{LB}}} p_i\right) / N_s^{\mathrm{LB}} \quad$ for $i = 1, \dots, N_s^{\mathrm{LB}}$

Set the power of the inactive streams to zero:

  $p_i \leftarrow 0 \quad$ for $i = N_s^{\mathrm{LB}} + 1, \dots, N_s$

Return p_i for $i = 1, \dots, N_s$

Appendix B

Extended Calculations

Bibliography

[1] G. Foschini and M. Gans, "On limits of wireless communications in a fading environment when using multiple antennas," *Wireless Personal Communications*, vol. 6, no. 3, pp. 311–335, Mar. 1998.

[2] S. Alamouti, "A simple transmit diversity technique for wireless communications," *IEEE Journal on Selected Areas in Communications*, vol. 16, no. 8, pp. 1451–1458, 1998.

[3] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[4] E. Telatar, "Capacity of multi-antenna gaussian channels," *European Transactions on Telecommunications*, vol. 10, no. 6, pp. 585–595, 1999.

[5] S. Vishwanath, N. Jindal, and A. Goldsmith, "On the capacity of multiple input multiple output broadcast channels," in *2002 IEEE International Conference on Communications. Conference Proceedings. ICC 2002 (Cat. No.02CH37333)*, vol. 3, 2002, pp. 1444–1450 vol.3.

[6] P. Viswanath and D. Tse, "Sum capacity of the vector gaussian broadcast channel and uplink-downlink duality," *IEEE Transactions on Information Theory*, vol. 49, no. 8, pp. 1912–1921, 2003.

[7] W. Yu and J. Cioffi, "Sum capacity of gaussian vector broadcast channels," *IEEE Transactions on Information Theory*, vol. 50, no. 9, pp. 1875–1892, 2004.

[8] S. Cornelis, "Optimization of linear hybrid beamformers on riemannian manifolds," Ph.D. dissertation, Ghent University, 2025.

[9] G. Strang, "The singular value decomposition (svd) of a matrix," 2016, MIT 18.095 lecture notes, Lecture 2.

[10] R. G. Gallager, *Information Theory and Reliable Communication.* John Wiley & Sons, 1968.

[11] T. M. Cover and J. A. Thomas, *Entropy, Relative Entropy, and Mutual Information.* John Wiley and Sons, Ltd, 2005, ch. 2, pp. 13–55.

[12] S. Boyd and L. Vandenberghe, *Convex Optimization.* Cambridge University Press, 2004.

[13] J. B. Lasserre, "On representations of the feasible set in convex optimization," 2009.